MATH 345 Skeleton Notes

Unit 2: Anatomy of a linear map

Section 2.4: Independence, bases, and rank

Linear independence

Activity

In the activity we showed that the set

$$W = \left\{ \begin{bmatrix} r \\ s \\ r + 2s \end{bmatrix} : r, s \in \mathbb{R} \right\}$$

is a subspace. In the activity, we showed that the set

$$S = \left\{ \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix} \right\}$$

is a spanning set for W. The set

$$T = \left\{ \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 2 \\ 0 \\ 2 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 3 \end{bmatrix} \right\}$$

also spans W. But why is S ‘better’?

A subspace U has many different spanning sets. We use the concept of *linear independence* to determine when sets of spanning vectors are the 'most efficient'. In data terms, independence asks whether one feature direction is already built from the others.

Definition: Linear Independence A set of vectors $\{\mathbf{x}_1, \dots, \mathbf{x}_k\}$ in $\mathbb{R}^n$ is said to be

__________, or simply __________

if the only solution to the equation $t_1\mathbf{x}_1 + \cdots + t_k\mathbf{x}_k = 0$ is trivial, i.e., $t_1 = \cdots = t_k = 0$.

A set of vectors $\{\mathbf{x}_1, \dots, \mathbf{x}_k\}$ in $\mathbb{R}^n$ is

__________ if it is *not* linearly independent, i.e., if there exists scalars $t_1, \dots, t_k$, not all zero, such that $t_1\mathbf{x}_1 + \cdots + t_k\mathbf{x}_k = 0$.

Activity

Determine whether the vectors

$$\mathbf{v}_1 = \begin{bmatrix} 3 \\ 2 \\ 2 \end{bmatrix} \quad \mathbf{v}_2 = \begin{bmatrix} 1 \\ 3 \\ 0 \end{bmatrix} \quad \mathbf{v}_3 = \begin{bmatrix} -1 \\ 4 \\ -2 \end{bmatrix}$$

are linearly independent.

The last example demonstrates an algorithm to verify whether a given set of vectors is linearly independent.
To verify if a set $\{\mathbf{x}_1, \dots, \mathbf{x}_k\}$ are linearly independent:

1. Consider the system of linear equations $t_1\mathbf{x}_1 + \cdots + t_k\mathbf{x}_k = 0$.

2. Solve the resulting linear homogeneous system.

3. If there is a nontrivial solution to the equation, then the vectors are linearly dependent. Otherwise, the vectors are linearly independent.

In matrix form: put the vectors $\mathbf{x}_1, \dots, \mathbf{x}_k$ as the columns of a matrix M. Then $\{\mathbf{x}_1, \dots, \mathbf{x}_k\}$ is independent exactly when $M\mathbf{c} = \mathbf{0}$ has only the trivial solution, equivalently when every column of M is a pivot column.

Theorem If $\{\mathbf{x}_1, \dots, \mathbf{x}_k\}$ is a linearly independent set of vectors in $\mathbb{R}^n$, then every vector in their span $\operatorname{span}\{\mathbf{x}_1, \dots, \mathbf{x}_k\}$ has a unique representation as a linear combination of the $\mathbf{x}_i$.

Proof Let $\mathbf{x}$ be an element of $\operatorname{span}\{\mathbf{x}_1, \dots, \mathbf{x}_k\}$. If the equation

$$\mathbf{x} = a_1\mathbf{x}_1 + \cdots + a_k\mathbf{x}_k$$

had another solution, i.e.,

$$\mathbf{x} = b_1\mathbf{x}_1 + \cdots + b_k\mathbf{x}_k,$$

then subtracting one equation from the other gives that

$$(a_1 - b_1)\mathbf{x}_1 + \cdots + (a_k - b_k)\mathbf{x}_k = 0.$$

Thus the coefficients $a_1 - b_1, \dots, a_k - b_k$ give a solution to the equation

$$t_1\mathbf{x}_1 + \cdots + t_k\mathbf{x}_k = 0.$$

But the vectors $\mathbf{x}_1, \dots, \mathbf{x}_k$ are linearly independent, and so this equation has only a trivial solution. Thus $a_1 - b_1, \dots, a_k - b_k = 0$, i.e., $a_1 = b_1, \dots, a_k = b_k$, proving that the representation of $\mathbf{x}$ as a linear combination is unique.

Activity

Determine whether the vectors

$$\mathbf{v}_1 = \begin{bmatrix} 1 \\ 2 \\ 0 \\ 1 \end{bmatrix} \quad \mathbf{v}_2 = \begin{bmatrix} 1 \\ -2 \\ 0 \\ 0 \end{bmatrix} \quad \mathbf{v}_3 = \begin{bmatrix} 0 \\ -3 \\ -1 \\ -1 \end{bmatrix}$$

are linearly independent.

Activity

Is the set $S = \{\mathbf{0}\}$ linearly independent?
Does the equation $t\mathbf{0} = \mathbf{0}$ have a nontrivial solution?

Fact The nonzero rows of a matrix in row echelon form are linearly independent.

Basis and dimension

In geometry, we often describe lines as being '1-dimensional', and planes as being '2-dimensional'. We can describe the notion of dimension more precisely using linear algebra.

Theorem: A Fundamental Theorem Let U be a subspace of $\mathbb{R}^n$. If U is spanned by m vectors, and if U contains k linearly independent vectors, then $k \leq m$.

Activity

Is it possible to have 4 linearly independent vectors in $\mathbb{R}^3$?

Definition: Bases of Vector Spaces If U is a subspace of $\mathbb{R}^n$, a set of vectors $\{\mathbf{x}_1, \ldots, \mathbf{x}_m\}$ in U is called a $\boxed{}$ of U if it satisfies the following two conditions:

- $\{\mathbf{x}_1, \ldots, \mathbf{x}_m\}$ is *linearly independent*.
- $U = \operatorname{span}\{\mathbf{x}_1, \ldots, \mathbf{x}_m\}$.

Theorem: The Invariance Theorem If $\{\mathbf{v}_1, \ldots, \mathbf{v}_m\}$ and $\{\mathbf{w}_1, \ldots, \mathbf{w}_k\}$ are both *bases* for a subspace U of $\mathbb{R}^n$, then $m = k$.

The Invariance Theorem says that, even though a subspace has many different bases, all bases must have the *same number* of vectors.

Definition: Dimensions of Vector Spaces The $\boxed{}$ of a non-zero subspace U of $\mathbb{R}^n$ is the *number* of vectors in a *basis* for U. We often write $\dim(U)$ for the dimension of U, and if $\dim(U) = d$, we say U is d-dimensional. By convention, the dimension of the trivial subspace $\{\mathbf{0}\}$ is *zero*.

- **Term:** spanning set
  Meaning: enough directions to generate the whole subspace
- **Term:** independent set
  Meaning: no direction is redundant
- **Term:** basis
  Meaning: just enough directions: spanning and independent
- **Term:** dimension
  Meaning: number of directions in any basis

Activity

What is the dimension of $\mathbb{R}^3$?

Activity

What is the dimension of $\mathbb{R}^4$?

Activity

What is the dimension of $\mathbb{R}^n$?

Activity

Find the dimension of the space

$$W = \left\{ \begin{bmatrix} r \\ s \\ r + 2s \end{bmatrix} : r, s \in \mathbb{R} \right\}.$$

Fact Let U be a non-zero subspace of $\mathbb{R}^n$. Then

- U has a basis and $\dim(U) \leq n$.
- Any independent set in U can be enlarged (by adding some subset of vectors in any fixed basis) to obtain a basis of U.
- Any spanning set for U contains a subset of vectors which form a basis for U.

Theorem Let U be a subspace of $\mathbb{R}^n$ where $\dim(U) = m$ and let $B = \{\mathbf{x}_1, \ldots, \mathbf{x}_m\}$ be a set of m vectors in U. Then B is linearly independent if and only if B spans U.

Proof Suppose B is linearly independent and $|B| = m = \dim(U)$. By A Fundamental Theorem, one cannot add any new vectors to the set B and remain linearly independent. But the referenced item says that if B is not already a spanning set for U, one can add new vectors to B from U, and remain linearly independent. Thus B must be a basis.
Conversely, if B spans U, then by the referenced item we can choose a subset of B which is a basis for U. But any proper subset of B contains fewer than m vectors, and thus by The Invariance Theorem, cannot be a basis for U. Thus B is the only subset of itself that can be a basis, and because B must contain a basis, the set B must itself be a basis.

Theorem Let U and W be subspaces of $\mathbb{R}^n$ with $U \subseteq W$. Then:

- $\dim(U) \leq \dim(W)$.
- If $\dim(U) = \dim(W)$, then $U = W$.

Activity

If $\mathbf{v}$ and $\mathbf{w}$ are nonzero vectors in $\mathbb{R}^3$, show that $\{\mathbf{v}, \mathbf{w}\}$ is dependent if and only if $\mathbf{v}$ and $\mathbf{w}$ are parallel, i.e., if one vector is a scalar multiple of the other.

Activity

Let $\mathbf{u}$, $\mathbf{v}$, and $\mathbf{w}$ are non-zero vectors in $\mathbb{R}^n$, where $\{\mathbf{v}, \mathbf{w}\}$ is independent. Show that $\{\mathbf{u}, \mathbf{v}, \mathbf{w}\}$ is independent if and only if $\mathbf{u}$ is not in the subspace $M = \operatorname{span}\{\mathbf{v}, \mathbf{w}\}$. See textbook figure fig-5-2-independent and textbook figure fig-5-2-dependent.

Note More generally, if $\{\mathbf{x}_1, \ldots, \mathbf{x}_k\}$ are a linearly independent set of vectors in $\mathbb{R}^n$, and $\mathbf{x}_{k+1}$ is another vector in $\mathbb{R}^n$, then $\{\mathbf{x}_1, \ldots, \mathbf{x}_{k+1}\}$ is a linearly independent set if and only if $\mathbf{x}_{k+1}$ is not contained in the span of the set $\{\mathbf{x}_1, \ldots, \mathbf{x}_k\}$.

Activity

Find a basis for $\mathbb{R}^4$ that contains the two vectors

$$\mathbf{v}_1 = \begin{bmatrix} 1 \\ 0 \\ 2 \\ 0 \end{bmatrix} \quad \text{and} \quad \mathbf{v}_2 = \begin{bmatrix} 0 \\ 0 \\ 3 \\ 3 \end{bmatrix}.$$

Bases and coordinate systems

One way to think of a basis is as providing a coordinate system for a subspace. The uniquely determined coefficients in the linear combination of a vector in the subspace are the coordinates.

Definition: Ordered Bases of a Subspace Let $S = \{\mathbf{b}_1, \ldots, \mathbf{b}_n\}$ be a basis of the subspace U. We call the tuple $B = (\mathbf{b}_1, \dots, \mathbf{b}_n)$ an of U.

As an example, $(\mathbf{e}_1, \mathbf{e}_2)$ and $(\mathbf{e}_2, \mathbf{e}_1)$ are two different ordered bases of $\mathbb{R}^2$. The two sets $\{\mathbf{e}_1, \mathbf{e}_2\}$ and $\{\mathbf{e}_2, \mathbf{e}_1\}$, and so describe the same bases of $\mathbb{R}^2$.

Note Many Linear Algebra textbooks, including *Linear Algebra With Applications, Nicholson* write ordered bases using set notation $\{\cdots\}$, despite the order in which the basis elements appear being important. We use tuple notation $(\cdots)$ to emphasize the importance of the order.

Given an ordered basis B for an m-dimensional subspace U of $\mathbb{R}^n$, and any vector $\mathbf{v} \in U$, we can describe its 'coordinates' with respect to the basis B, identifying $\mathbf{v}$ with a vector in $\mathbb{R}^m$.

Definition: Coordinate Vectors With Respect to a Basis Let U be an m-dimensional subspace of $\mathbb{R}^n$, and let $B = (\mathbf{b}_1, \dots, \mathbf{b}_m)$ be an ordered basis for U. Given any vector $\mathbf{v} \in U$, there exists unique scalars $t_1, \dots, t_m$ such that

$$\mathbf{v} = t_1\mathbf{b}_1 + \cdots + t_m\mathbf{b}_m$$

and we define the of a vector $\mathbf{v} \in U$ with respect to B to be the m-vector

$$C_B(\mathbf{v}) = \begin{bmatrix} t_1 \\ \vdots \\ t_m \end{bmatrix}.$$

See textbook figure fig-5-2-coordinate-basis-diagram for an illustration of this concept, taken from *Stanford's MATH 51 Textbook*.

Note If $B = (\mathbf{b}_1, \dots, \mathbf{b}_m)$ is an ordered basis for a subspace U, then $C_B(\mathbf{b}_i) = \mathbf{e}_i$, where $\mathbf{e}_\mathrm{i}$ is the ith standard basis vector in $\mathbb{R}^m$.

Activity

Let $B = (\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3)$ be an ordered basis for $\mathbb{R}^3$, where

$$\mathbf{v}_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \quad \mathbf{v}_2 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} \quad \mathbf{v}_3 = \begin{bmatrix} 2 \\ 0 \\ 1 \end{bmatrix}.$$

What is $C_B(\mathbf{v})$ if

$$\mathbf{v} = \begin{bmatrix} 1 \\ 1 \\ -5 \end{bmatrix}.$$

Rank of a matrix

We are now going to connect the concept of dimension with the notion of rank. In this section, we will often realize vectors in $\mathbb{R}^n$ as rows rather than columns. The notion of span, linear independence, and basis are defined analogously as how they are defined for column vectors.

Definition: Column and Row Space Let

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix}$$

be an $m \times n$ matrix.

- The columns of A, considered as vectors in $\mathbb{R}^m$ span a *subspace* of $\mathbb{R}^m$ called the $\boxed{\qquad\qquad\qquad}$ of A, denoted $\mathrm{col}(A)$.
- The rows of A, considered as vectors in $\mathbb{R}^n$, span a *subspace* of $\mathbb{R}^n$ called the $\boxed{\qquad\qquad}$ of A, denoted $\mathrm{row}(A)$.

Note The column space is the image space from Subspaces written with a column-based name:

$$\mathrm{col}(A) = \mathrm{im}(A).$$

The notation $\mathrm{im}(A)$ emphasizes outputs of the map $\mathbf{x} \mapsto A\mathbf{x}$. The notation $\mathrm{col}(A)$ emphasizes the columns that span those outputs.

Lemma Let A and B be two row equivalent $m \times n$ matrices. Then $\mathrm{row}(A) = \mathrm{row}(B)$.

Proof Each time we apply a row operation to a matrix, we replace one or more of its rows with a linear combination of other rows. Thus the row space of a matrix A' obtained by applying a row operation to a matrix A must be contained in the row space of A. But applying the argument in reverse (since every row operation is reversible), the row space of A must be contained in the row space of A'. Thus the two row spaces are equal. But if the row space is not changed after applying a single row operation, it is not changed after applying an arbitrary number of row operations, and so the result follows.

Lemma If R is a matrix in row echelon form (recall Row echelon form and reduced row echelon form) then the non-zero rows of R are a basis of $\mathrm{row}(R)$.

Proof The nonzero rows of R are linearly independent by the referenced item, and span $\mathrm{row}(R)$ by definition. They thus by definition form a basis for $\mathrm{row}(R)$.

Remark In Rank of a matrix, we defined the *rank* of a matrix as the number of leading 1s (i.e., the number of pivots) of any matrix R in row echelon form which is row-equivalent to A. By the referenced item, $\mathrm{rank}(A) = \dim(\mathrm{row}(A))$, which justifies that the rank of A is independent of the choice of matrix R.

Activity

Find a basis for $V = \mathrm{span}\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \mathbf{v}_4\}$, where

$$\mathbf{v}_1 = \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix} \quad \mathbf{v}_2 = \begin{bmatrix} 1 \\ 2 \\ -3 \end{bmatrix} \quad \mathbf{v}_3 = \begin{bmatrix} 0 \\ 1 \\ -2 \end{bmatrix} \quad \mathbf{v}_4 = \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix}.$$

Theorem: Rank Theorem Let A denote any $m \times n$ matrix of rank r. Then

$$\dim(\mathrm{col}(A)) = \dim(\mathrm{row}(A)) = r.$$

Moreover, if A is row-equivalent to a matrix R in row echelon form, then

- The r leading rows of R are a basis for $\mathrm{row}(A)$.
- If the pivots of R lie in columns $j_1, j_2, \dots, j_r$, then the columns $j_1, j_2, \dots, j_r$ of A are a basis for $\mathrm{col}(A)$.

Warning: Pivot columns come from the original matrix Row reduction identifies pivot positions, but row operations do not preserve the original columns of A. When finding a basis for $\mathrm{col}(A)$, use the pivot columns of the original matrix A, not the columns of the row-reduced matrix.

- **Goal:** basis for $\mathrm{row}(A)$
  Row reduction tells you: nonzero rows of REF/RREF
  Use these vectors: rows of the reduced matrix
- **Goal:** basis for $\mathrm{col}(A)$
  Row reduction tells you: pivot column positions
  Use these vectors: corresponding columns of original A
- **Goal:** basis for $\mathrm{null}(A)$
  Row reduction tells you: free variables
  Use these vectors: basic solution vectors in $\mathbb{R}^n$
- **Goal:** rank
  Row reduction tells you: number of pivots
  Use these vectors: a number
- **Goal:** nullity
  Row reduction tells you: number of free variables
  Use these vectors: a number

Fact It follows from Rank Theorem that for any matrix A, $\mathrm{rank}(A) = \mathrm{rank}(A^T)$.

Activity

Let

$$A = \begin{bmatrix} 2 & 1 & 0 \\ 1 & 2 & -3 \\ 0 & 1 & -2 \\ 1 & 1 & -1 \end{bmatrix}$$

be the matrix we considered in the solution to the activity.
What is $\dim(\mathrm{row}(A))$?

Activity

Let

$$A = \begin{bmatrix} 2 & 1 & 0 \\ 1 & 2 & -3 \\ 0 & 1 & -2 \\ 1 & 1 & -1 \end{bmatrix}$$

be the matrix we considered in the solution to the activity.
Find a basis for the column space of A.

Activity

Let

$$A = \begin{bmatrix} 2 & 1 & 0 \\ 1 & 2 & -3 \\ 0 & 1 & -2 \\ 1 & 1 & -1 \end{bmatrix}$$

be the matrix we considered in the solution to the activity.
What is $\dim(\mathrm{col}(A))$?

Activity

Let

$$A = \begin{bmatrix} 2 & 1 & 0 \\ 1 & 2 & -3 \\ 0 & 1 & -2 \\ 1 & 1 & -1 \end{bmatrix}$$

be the matrix we considered in the solution to the activity.
Find a basis for the nullspace of A.

Definition The $\boxed{}$ of a matrix A is equal to $\dim(\mathrm{null}(A))$. It is denoted $\mathrm{nullity}(A)$.

Rank counts independent directions transmitted by a matrix. Nullity counts independent input directions forgotten by a matrix. The row-reduction definition of rank now connects to dimension.

Theorem: Rank-Nullity Theorem Let A denote an $m \times n$ matrix of rank r. Then

- The $n - r$ basic solutions to the equation $A\mathbf{x} = \mathbf{0}$ provided by the Gaussian algorithm are a basis for $\mathrm{null}(A)$, so $\dim(\mathrm{null}(A)) = n - r$.
- $\dim(\mathrm{im}(A)) = r$.
- $\dim(\mathrm{im}(A)) + \dim(\mathrm{null}(A)) = n$.
- $\mathrm{rank}(A) + \mathrm{nullity}(A) = n$.

Note

input dimensions = transmitted dimensions + forgotten dimensions

$$n = \mathrm{rank}(A) + \mathrm{nullity}(A)$$

We can now state one of the preview patterns from Section 2.1, introduced in Two inputs, one output, in the language of null spaces.

Activity: Same output from a null-space direction

Suppose $A\mathbf{x} = \mathbf{y}$ and $A\mathbf{z} = \mathbf{0}$. Compute $A(\mathbf{x} + t\mathbf{z})$, where $t \in \mathbb{R}$. What does this say about uniqueness?

Activity: Revisiting the height map with rank and nullity

Return again to A map that forgets height. Let

$$P = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}.$$

Find the column space, null space, rank, and nullity of P. How does rank-nullity describe the height direction that P forgets?

Activity

Consider the matrix

$$B = \begin{bmatrix} 3 & -1 & 2 & 0 \\ -6 & 2 & -2 & 6 \end{bmatrix}.$$

You may use the fact that the reduced row echelon form of B is the matrix

$$\begin{bmatrix} 1 & -1/3 & 0 & -2 \\ 0 & 0 & 1 & 3 \end{bmatrix}.$$

Find a basis for $\mathrm{im}(B)$.

Activity

Consider the matrix

$$B = \begin{bmatrix} 3 & -1 & 2 & 0 \\ -6 & 2 & -2 & 6 \end{bmatrix}.$$

You may use the fact that the reduced row echelon form of B is the matrix

$$\begin{bmatrix} 1 & -1/3 & 0 & -2 \\ 0 & 0 & 1 & 3 \end{bmatrix}.$$

Find a basis for $\mathrm{null}(B)$.

Redundant features in a data matrix

A data matrix can have redundant columns. This means one feature column is a linear combination of other feature columns.
Rows below represent houses. The three features are

first-level area, second-level area, total area,

measured in hundreds of square feet:

$$X = \begin{bmatrix} 9 & 7 & 16 \\ 11 & 9 & 20 \\ 14 & 0 & 14 \\ 8 & 8 & 16 \end{bmatrix}.$$

Let $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3$ be the columns of X. The third column satisfies

$$\mathbf{v}_3 = \mathbf{v}_1 + \mathbf{v}_2,$$

so the third feature carries no new independent direction. The dependence relation is

$$\mathbf{v}_1 + \mathbf{v}_2 - \mathbf{v}_3 = \mathbf{0},$$

or equivalently

$$X \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix} = \mathbf{0}.$$

The null-space direction

$$\mathbf{z} = \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix}$$

says that coefficient vectors differing by a multiple of $\mathbf{z}$ give the same prediction vector.

Activity: Redundant square-footage features

Use the data matrix

$$X = \begin{bmatrix} 9 & 7 & 16 \\ 11 & 9 & 20 \\ 14 & 0 & 14 \\ 8 & 8 & 16 \end{bmatrix}, \qquad \mathbf{z} = \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix}.$$

1. Compute $X\mathbf{z}$.
2. What does this say about the three feature columns?
3. Compare the coefficient vectors

$$\mathbf{c} = \begin{bmatrix} 3 \\ 1 \\ 0 \end{bmatrix}, \qquad \mathbf{c}' = \begin{bmatrix} 2 \\ 0 \\ 1 \end{bmatrix}.$$

   Compute $X\mathbf{c}$ and $X\mathbf{c}'$.
4. The two coefficient vectors tell different stories. Why do they make the same predictions?
5. What warning does this give about asking which feature is most important?

Note: Common exact feature redundancies Common exact feature redundancies include:

- first-level area, second-level area, and total area;
- minutes and seconds, where seconds $= 60 \cdot$ minutes;
- height in inches and height in centimeters, where centimeters $=$ $2.54 \cdot$ inches;
- subtotal, tax, tip, and total bill, where total $=$ subtotal $+$ tax $+$ tip;
- exam parts and exam total, where total $=$ part $1 +$ part $2 +$ part 3.

These are linear redundancies. Not every relationship between features is linear. For example, rectangle area is width times height, not a linear combination of width and height.

Activity: Which features are linearly redundant?

For each feature list, decide whether one feature is a linear combination of the others.

1. first-level area, second-level area, total area;
2. height in inches, height in centimeters;
3. width, height, area of a rectangle;
4. quiz 1 score, quiz 2 score, quiz total;
5. bedrooms, bathrooms, selling price.

Activity: Auditing redundant features in code

The matrix below stores the house features from the previous activity.

```
import numpy as np

X = np.array([
    [9, 7, 16],
    [11, 9, 20],
    [14, 0, 14],
    [8, 8, 16],
])

z = np.array([1, 1, -1])
c = np.array([3, 1, 0])
c_alt = np.array([2, 0, 1])

rank = int(np.linalg.matrix_rank(X))

X @ z, rank, X @ c, X @ c_alt
```

Here `c_alt` stores the vector $\mathbf{c}'$.
Output:

```
(array([0, 0, 0, 0]), 2, array([34, 42, 42, 32]), array([34
```

1. What does the zero output mean?
2. What does the rank say about the three feature columns?
3. Why do $X\mathbf{c}$ and $X\mathbf{c}'$ agree?
4. What does this say about interpreting coefficients when features are redundant?

Activity: A difference matrix forgets level

Let

$$D = \begin{bmatrix} -1 & 1 & 0 & 0 \\ 0 & -1 & 1 & 0 \\ 0 & 0 & -1 & 1 \end{bmatrix}.$$

Compute $D\begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \end{pmatrix}$. What input direction is forgotten?

Activity: Reading the difference matrix in code

```
            D = np.array([
    [-1, 1, 0, 0],
    [0, -1, 1, 0],
    [0, 0, -1, 1],
])

x = np.array([2, 5, 9, 10])
ones = np.ones(4, dtype=int)

D @ x, D @ (x + 10*ones), D @ ones
```

Output:

```
            (array([3, 4, 1]), array([3, 4, 1]), array([0, 0, 0]))
```

1. What does `D @ x` measure?
2. Why do the first two outputs agree?
3. What does `D @ ones` say about the map?

Redundant features create null-space directions. If $X\mathbf{z} = \mathbf{0}$, then

$$X(\mathbf{c} + t\mathbf{z}) = X\mathbf{c}$$

for every scalar t. Thus many coefficient vectors can produce the same prediction vector. Rank counts how many independent feature directions remain; nullity counts how many independent coefficient changes are invisible to the data matrix.